

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and trained **deep learning networks**, applying state-of-the-art computer vision techniques to automotive domain challenges, contributing to next-generation automated driving systems.
- Selected and refined **appropriate datasets and data representation methods** for deep learning model training, enhancing scene understanding capabilities by **15%** in object classification accuracy.
- Collaborated with data engineers and analysts to implement robust **data acquisition and data integrity** protocols, ensuring optimal data pipelines for model development and validation.
- Executed **statistical analysis and fine-tuning** based on test results, improving model performance metrics by an average of **12%** across key benchmarks.
- Managed **Linux environments** for research and development, including shell scripting for automation and log analysis, supporting a team of 5 researchers.
- Contributed to the development of **software for embedded platforms** by optimizing algorithms and performing integration testing, ensuring seamless deployment of DL models.

SKILLS

Machine Learning – Deep Learning, Keras, PyTorch, scikit-learn, statistical analysis, regression, hypothesis testing, machine learning fundamentals.

Computer Vision – 3D object detection, classification, structure-from-motion, localization & mapping, OpenCV.

Programming & Scripting – Python, Java, R, SQL, Bash scripting.

Data Management – Data acquisition, data integrity, data pipelines, data representation, data modelling, data warehousing.

Development Tools – Git, Jenkins, Docker, Linux.

Frameworks & Libraries – Spring Boot, pandas, NumPy, Flask, FastAPI.

Cloud & Platforms – AWS (EC2, S3, Lambda, CloudWatch), embedded platforms.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Automotive Pedestrian Detection & Classification System (MSc Project)

- Designed and implemented a **deep learning network** using **PyTorch** and **OpenCV** for real-time pedestrian detection and classification in complex urban driving scenarios.
- Processed and curated a large-scale dataset of automotive imagery, achieving **98.5% accuracy** in identifying pedestrians under varying lighting and weather conditions.

- Optimized model inference speed for embedded automotive platforms, reducing latency by **30%** while maintaining high detection precision through **statistical analysis** and fine-tuning.

HCL Payments Operations Dashboard Enhancements

- Developed **Java Spring Boot** microservices to enhance transaction monitoring capabilities on a financial-services dashboard, processing over **1 million transactions daily**.
- Automated data validation and reconciliation workflows using **Python scripting** and **SQL** queries across PostgreSQL and Oracle databases, reducing manual processing time by **20%**.
- Implemented production support improvements for the dashboard, responding to P1/P2 incidents within SLA targets and contributing to **99.9% uptime** for critical financial operations.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)